ARIA-GOLF v1–v5: Structural Differences and Model Equations

Internal architecture note

June 13, 2026

Abstract

This note reorganizes the v1–v5 ARIA-GOLF configuration differences into a model-comparison form. All versions share the same encoder family, observed F0 conditioning, LF glottal-flow oscillator, white-noise generator, room filter, and multi-scale spectrogram loss. The relevant architectural differences are localized to two places: the analytic vocal-tract cascade (v1–v3) and the harmonic/noise excitation topology (v3–v5). In all versions, the harmonic and noise branches are first combined into one source signal s, and this combined source is then passed through the vocal tract once. The equations below make these changes explicit so that the five variants can be presented as an interpretable ablation sequence.

How to Read the Five Versions

The simplest reading is that v1–v3 are not five unrelated systems. They are an ablation path with two axes. First, v1–v3 refine the analytic vocal tract: the model searches for a useful balance between explicit formant control, bandwidth constraints, spectral tilt, and learned residual all-pole capacity. Second, v4 and v5 keep the v3 vocal tract and change only the excitation mixture, because the remaining artifact is caused by additive harmonic-plus-noise synthesis.

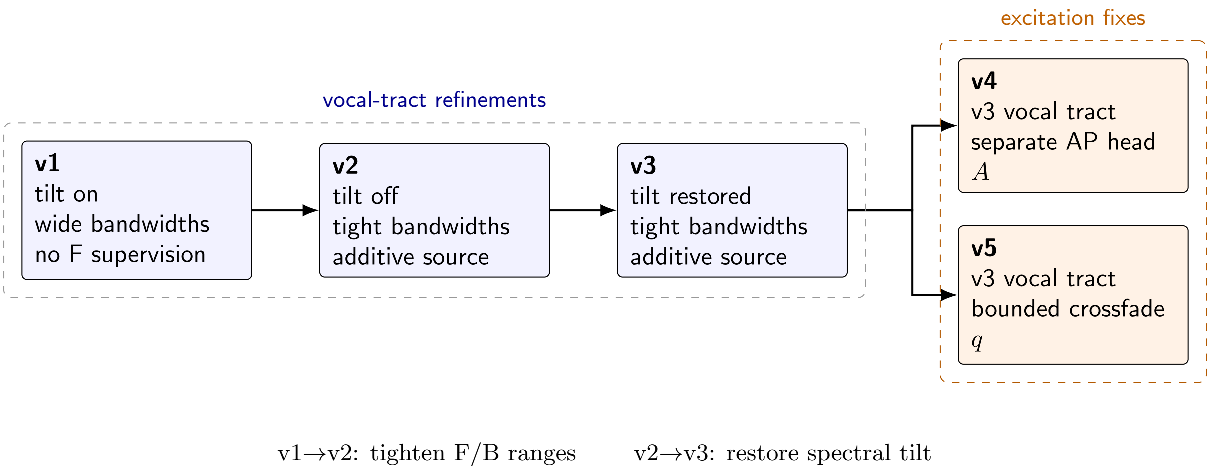

Figure 1: Conceptual lineage of ARIA-GOLF v1–v5. v1–v3 should be read as vocal-tract refinements; v4 and v5 are two alternative fixes for the same high-band excitation artifact. v1 is also the unsupervised-formant baseline.

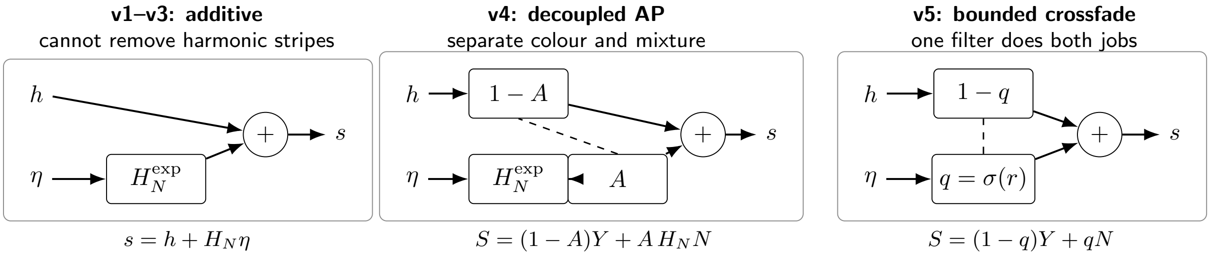

Figure 2: Excitation topologies. Additive synthesis can add noise but cannot suppress periodic harmonics. v4 and v5 introduce a per-band harmonic/noise mixture, so high-frequency periodic energy can be converted into shaped noise.

1 Shared Source-Filter Skeleton

Let x denote the input waveform and let $F_{0,k}$ be the observed frame-rate pitch contour. The encoder predicts frame-rate controls

$$\mathbf{c}_k = E_\theta(x, F_{0,k}), \tag{1}$$

where the final projection width is determined by the decoder's declared control splits. After upsampling, the LF source phase is obtained by

$$\phi_n = \phi_{n-1} + 2\pi \frac{F_{0,n}}{f_\mathrm{s}}, \tag{2}$$

and the deterministic harmonic source is

$$h_n = v_n\, g_{\mathrm{LF}}(\phi_n; R_{d,n}), \tag{3}$$

where $v_n = \mathbb{1}[F_{0,n} > 0]$ is the voicing gate and R_d controls the LF glottal-flow shape. A white-noise excitation $\eta_n \sim \mathcal{N}(0, 1)$ is available in every version. Each model first combines the harmonic and noise branches into a single source/excitation signal $s^{(m)}$. The final waveform is always

$$\hat{x} = H_{\mathrm{room}} * \left(H_{\mathrm{VT}}^{(m)} * s^{(m)}\right), \tag{4}$$

where $m \in \{1, \ldots, 5\}$ indexes the model version. Thus each version differs only in the version-specific source $s^{(m)}$ and vocal-tract filter $H_{\mathrm{VT}}^{(m)}$. In implementation this is one vocal-tract pass over the combined source, i.e., LPC(h + noise), not two separately filtered branches that are summed afterward. Since the end filter is linear, these are mathematically equivalent, but the source-level ordering is the clearest way to read the architecture.

2 Vocal-Tract Cascade

The analytic vocal tract consists of a gain term, an optional spectral-tilt term, two deterministic all-pole formant sections, and a set of learned biquad sections:

$$H_{\mathrm{VT}}{}_n^{(m)}(z) = G_n\, T_{\alpha,n}^{\delta_m}(z) \prod_{j=1}^{2} P(z; F_{j,n}, B_{j,n}) \prod_{\ell=1}^{L_m} B_{\ell,n}(z). \tag{5}$$

Here $\delta_m = 1$ when spectral tilt is enabled and $\delta_m = 0$ otherwise. The learned residual sections $B_{\ell,n}$ absorb vocal-tract detail not captured by the explicitly controlled F1/F2 poles. The table below reports the 16 kHz F024 configurations, where v4/v5 were defined.

Version	Tilt δ_m	Learned L_m	F1 range	F2 range	B1/B2 range	LPC order
v1	1	6	150–1300	600–3200	30–330 / 30–430	18
v2	0	7	200–1000	700–3000	50–200 / 50–250	18
v3	1	6	150–1300	600–3200	50–200 / 50–250	18
v4	1	6	150–1300	600–3200	50–200 / 50–250	18
v5	1	6	150–1300	600–3200	50–200 / 50–250	18

Table 1: Vocal-tract differences across v1–v5 in the 16 kHz F024 configs. v4 and v5 inherit the v3 vocal-tract design; their changes are in the excitation topology.

The nominal LPC order is counted as twice the number of biquad-like sections:

$$O_m = 2\big(\delta_m + 2 + L_m\big). \tag{6}$$

Thus the 16 kHz v1/v3/v4/v5 vocal tracts have $2(1+2+6) = 18$, while v2 has $2(0+2+7) = 18$. The tilt section $T_\alpha(z)$ is implemented as a degenerate biquad with one real pole, so an effective pole count of 17 may appear in older comments for tilt-enabled 16 kHz models. This note uses the nominal section-count convention.

Sample rate / speaker	Nominal LPC order	v1 L_m	v2 L_m	v3 L_m
16 kHz / F024	18	6	7	6
22 kHz / LJSpeech	22	8	9	8
24 kHz / CSMSC	22	8	9	8

Table 2: Sample-rate dependence of the learned residual cascade. The analytic F1/F2 controls and spectral tilt stay conceptually fixed; higher sample rates use more learned biquads to cover additional high-frequency detail.

3 Version-Specific Equations

3.1 v1: First Analytic-Formant Baseline

v1 introduces the interpretable F1/F2 vocal-tract cascade with spectral tilt, wide bandwidth constraints, and six learned residual biquads:

$$H_{\mathrm{VT}}{}_n^{(1)}(z) = G_n T_{\alpha,n}(z) \prod_{j=1}^{2} P(z; F_{j,n}, B_{j,n}) \prod_{\ell=1}^{6} B_{\ell,n}(z), \tag{7}$$

$$s_n^{(1)} = h_n + (H_{\mathrm{N}}{}_n^{\exp} * \eta)_n. \tag{8}$$

The additive excitation can raise the noise floor but cannot attenuate coherent high-frequency harmonics already present in h_n. Importantly, the v1 F1/F2 controls are not explicitly supervised in the F024 training recipe; they emerge only through the reconstruction objective, which makes v1 the loosest formant-control baseline.

3.2 v2: Tighter Formant Control without Spectral Tilt

v2 removes the tilt branch, narrows the F1/F2 and B1/B2 constraints, and adds one extra learned biquad:

$$H_{\mathrm{VT}}{}_n^{(2)}(z) = G_n \prod_{j=1}^{2} P(z; F_{j,n}, B_{j,n}) \prod_{\ell=1}^{7} B_{\ell,n}(z), \tag{9}$$

$$s_n^{(2)} = h_n + (H_{\mathrm{N}}{}_n^{\exp} * \eta)_n. \tag{10}$$

The intent is stronger formant manipulability, but dropping tilt forces the residual sections to explain broad spectral slope.

3.3 v3: Current Vocal-Tract Base

v3 restores spectral tilt while keeping the tighter bandwidth constraints from v2 and returning to six learned biquads:

$$H_{\mathrm{VT}}{}_n^{(3)}(z) = G_n T_{\alpha,n}(z) \prod_{j=1}^{2} P(z; F_{j,n}, B_{j,n}) \prod_{\ell=1}^{6} B_{\ell,n}(z), \tag{11}$$

$$s_n^{(3)} = h_n + (H_{\mathrm{N}}{}_n^{\exp} * \eta)_n. \tag{12}$$

This is the strongest analytic vocal-tract baseline: it combines F1/F2 control, bounded bandwidths, spectral tilt, and residual all-pole capacity. Its main remaining weakness is the additive-only excitation, which can leave high-frequency harmonic stripes in strongly voiced regions.

3.4 v4: Decoupled Band-Aperiodicity Source

v4 keeps the v3 vocal tract but adds an explicit aperiodicity filter $A_n(\omega) \in [0, 1]$ with a rising high-frequency prior:

$$H_{\mathrm{VT}}{}_n^{(4)}(z) = H_{\mathrm{VT}}{}_n^{(3)}(z), \tag{13}$$

$$A_n(\omega) = \sigma\big(a_n(\omega) + b_{\mathrm{HF}}(\omega)\big), \tag{14}$$

$$S_n^{(4)}(\omega) = \big(1 - A_n(\omega)\big) Y_n(\omega) + A_n(\omega) H_{\mathrm{N}}{}_n^{\exp}(\omega) N_n(\omega). \tag{15}$$

Here $Y_n(\omega)$ and $N_n(\omega)$ are the short-time spectra of the harmonic source and white noise, respectively. The noise-colour filter H_{N} remains unbounded, while A separately controls the harmonic/noise mixture. This makes v4 the most principled fix for the high-band harmonic-stripe artifact.

3.5 v5: Minimal Bounded Crossfade

v5 also keeps the v3 vocal tract, but uses the noise filter itself as a bounded crossfade:

$$H_{\mathrm{VT}}{}_n^{(5)}(z) = H_{\mathrm{VT}}{}_n^{(3)}(z), \tag{16}$$

$$q_n(\omega) = \sigma(r_n(\omega)), \tag{17}$$

$$S_n^{(5)}(\omega) = \big(1 - q_n(\omega)\big) Y_n(\omega) + q_n(\omega) N_n(\omega). \tag{18}$$

Equivalently, in operator form,

$$s^{(5)} = (I - Q)h + Q\eta = h + Q\eta - Qh. \tag{19}$$

This is the minimal structural fix: no extra aperiodicity head is introduced, but the same bounded filter must serve both as noise colour and aperiodicity weight.

4 Interpretation as an Ablation Sequence

Version	Formant loss	Smoothness	Residual reg.	Batch $\times$ steps	Data passes
v1	0	–	–	64×40k	$1\times$
v2	1.5	–	0.02	64×80k	$2\times$
v3	2.0	0.05	0.02	128×40k	$2\times$
v4	2.0	0.05	0.02	128×40k	$2\times$
v5	2.0	0.05	0.02	128×40k	$2\times$

Table 3: Training and supervision differences in the F024 comparison. The main revision is that v1 has no explicit formant supervision; the earlier "light supervision" reading should not be used.

Version	Intent	Main change	Trade-off
v1	First analytic-formant ARIA	Tilt on, wide bandwidths, no formant supervision	Good reconstruction, loose/unconstrained formant control
v2	Stronger formant control	Tilt off, tighter bandwidths, seven learned sections	Control improves, spectral slope less explicit
v3	Current base	Tilt restored, tight bandwidths, strong supervision	Good manipulation, additive source artifact remains
v4	Principled artifact fix	Adds decoupled band-aperiodicity A	Most expressive, one extra control head
v5	Minimal artifact fix	Bounded noise filter becomes crossfade q	No extra head, but noise level and aperiodicity are coupled

Table 4: Recommended narrative for presenting v1–v5.

5 Evaluation Implications

The v1–v3 comparison should be evaluated mainly through formant-control error, copy-synthesis quality, and whether spectral tilt improves naturalness without weakening manipulation. The v3–v5 comparison should focus on high-frequency voiced regions, because the core hypothesis is that additive excitation cannot remove coherent high-band harmonics, whereas v4 and v5 can convert part of the harmonic source into shaped noise. The most informative visualizations are:

1. a lineage diagram showing v1–v3 as vocal-tract refinements and v4–v5 as excitation fixes;
2. a side-by-side excitation-flow figure for additive, decoupled aperiodicity, and bounded crossfade;
3. a spectrogram strip comparing v3, v4, and v5 in the 4–8 kHz band;
4. learned aperiodicity curves $A_n(\omega)$ for v4 and $q_n(\omega)$ for v5 on voiced frames.